

Module 9 Lab:

Corporate Wi-Fi Network Control with FreeRADIUS

JR Clark (002035589)

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Introduction

Enterprise wireless networks sit at the intersection of identity, security, and operational continuity. As organizations adopt Zero Trust principles and face increasing regulatory pressure from frameworks such as PCI DSS and GDPR, network access can no longer be treated as a flat, binary decision. Access must reflect who the user is, what they need to do, and when they are authorized to do it. FreeRADIUS provides a flexible, attribute-driven mechanism for enforcing these distinctions at the moment a device joins the network, making it well-suited for environments where identity-based access control is essential.

This lab implements a realistic corporate Wi-Fi access control model that differentiates between Full-Time Employees (FTEs), Contractors, and Guests. Each group represents a distinct combination of trust level, business need, and risk exposure. FTEs require continuous access to internal systems and long-running sessions; Contractors require limited, time-bounded access aligned with business hours; and Guests require temporary, internet-only connectivity tied to physical identity verification. These requirements translate into specific RADIUS authorization controls, including VLAN assignments for network segmentation, Login-Time restrictions for temporal access control, and Session-Timeout values to limit exposure.

The purpose of this lab is not only to configure FreeRADIUS, but to demonstrate how identity-based network segmentation can be operationalized through concrete RADIUS attributes. By implementing and testing these policies in a controlled Kali Linux environment, the lab shows how authentication and authorization decisions can be enforced consistently and transparently. The resulting configuration illustrates how FreeRADIUS functions as a foundational component of a broader Zero Trust architecture, complementing firewall, DLP, and AI governance controls that operate at higher layers of the security stack.

Scenario: Corporate Wireless Access Control Using FreeRADIUS

Modern enterprises increasingly rely on identity-driven network segmentation to enforce least-privilege access, reduce insider-risk exposure, and meet regulatory expectations such as PCI DSS and GDPR. In this environment, wireless access cannot be uniform across all users. Instead, access must reflect the user's role, the sensitivity of the systems they interact with, and the organization's operational and compliance requirements. This scenario models a corporate Wi-Fi deployment where FreeRADIUS provides centralized authentication and authorization for three distinct groups: Full-Time Employees (FTEs), Contractors, and Guests. Each group receives a different level of access based on business need and risk tolerance, enforced through RADIUS attributes such as VLAN assignment, Login-Time restrictions, and Session-Timeout values.

Organizational Context

The organization operates two wireless networks: a Corporate Wi-Fi network for employees and contractors, and a Guest Wi-Fi network for visitors. Both networks authenticate against FreeRADIUS, but the authorization outcomes differ significantly. Corporate Wi-Fi provides access to internal systems such as the intranet, SharePoint, Workday, FieldGlass, and engineering tools. Guest Wi-Fi provides internet-only access and is intentionally isolated to prevent data leakage or unauthorized lateral movement.

The company also enforces strict AI governance and data-handling policies. Public LLMs (e.g., ChatGPT, Gemini, Claude) are blocked for all users, and only the enterprise-sandboxed version of Microsoft Copilot is permitted for FTEs and Contractors. Uploading files to non-company cloud storage platforms (e.g., Google Drive, Dropbox, Box) is prohibited for all

user types. These controls are enforced through firewall and DLP systems and complement the network segmentation implemented through FreeRADIUS.

Group 1: Full-Time Employees (FTEs)

Business Requirements: FTEs require broad, continuous access to corporate systems. Their work may extend into evenings or weekends, especially during incident response or maintenance windows. They need stable, long-running sessions and unrestricted access to internal applications.

Authorization Policies

- **VLAN 10:** Full corporate network access.
- **24/7 Login-Time:** No time-of-day restrictions.
- **No session timeout:** Supports extended work sessions.
- **Access to internal systems:** Intranet, SharePoint, Workday, FieldGlass, engineering tools.
- **AI governance:** Blocked from public LLMs; permitted to use corporate-sandboxed CoPilot only.
- **Data loss prevention:** Blocked from uploading files to non-Company cloud storage.

These policies reflect the higher trust posture of FTEs and the operational need for continuous access.

Group 2: Contractors

Business Requirements: Contractors perform scoped project work and do not require access to internal corporate systems. They are not authorized to be onsite outside business hours and do not participate in after-hours support. Their access must be limited to reduce risk exposure and align with least-privilege principles.

Authorization Policies

- **VLAN 20:** Restricted corporate network segment with no routing to internal systems.
- **Session timeout: 1 hour:** Limits unattended or stale sessions.
- **Login-Time: 08:00–18:00, Monday–Friday:**
 - No on-site network login between 6pm and 8am.
 - Authentication attempts at 7pm, 11pm, or 2am result in **Access-Reject**.
- **Forced disconnect:** Sessions crossing into off-hours are terminated.
- **No access to internal systems:** Intranet, SharePoint, Workday, FieldGlass.
- **AI governance:** Blocked from public LLMs; permitted to use corporate-sandboxed CoPilot only.
- **Data loss prevention:** Blocked from uploading files to non-Company cloud storage.

These controls reduce after-hours exposure and limit contractors to the minimum access required for their work.

Group 3: Guests

Business Requirements: Guests include visitors, vendors, and interview candidates. They require temporary, internet-only access and must never reach internal systems or corporate AI tools. To prevent unauthorized after-hours access—such as individuals connecting from the parking lot—the Guest Wi-Fi is tightly controlled and tied to physical identity verification.

Guest Access Workflow

1. Guest signs in at the front desk and presents identification.
2. Guest signs a **non-disclosure agreement (NDA)**.
3. Guest receives the **password-of-the-day** for the Guest Wi-Fi.
4. When connecting, the guest must enter:

- Their **registered email address**,
- The **password-of-the-day**, and
- A **Terms of Use (TOU)** acceptance checkbox.

Authorization Policies

- **VLAN 30:** Internet-only network segment.
- **Session timeout: 30 minutes:** Limits exposure and prevents long-running sessions.
- **Login-Time: 08:00–18:00, Monday–Friday:**
 - Guest Wi-Fi **shuts down at 6pm.**
 - Prevents after-hours access from outside the building.
- **No access to internal systems.**
- **AI governance:** Blocked from all LLMs, including CoPilot.
- **Data loss prevention:** Blocked from uploading files to non-Company cloud storage.

This workflow binds physical identity to digital access and reduces the risk of unauthorized use.

FreeRADIUS Enforcement

These policies are implemented in FreeRADIUS through:

- **VLAN assignment** using Tunnel-Type, Tunnel-Medium-Type, and Tunnel-Private-Group-Id.
- **Session timeout** using Session-Timeout.
- **Time-of-day restrictions** using Login-Time.
- **Access-Reject** responses for unauthorized login attempts.

Firewall and DLP systems complement RADIUS by enforcing cloud-upload restrictions and AI governance controls. FreeRADIUS enforces identity-based network segmentation (VLANs),

session limits, and time-of-day restrictions. Additional controls such as AI governance and cloud-upload restrictions are enforced by firewall and DLP systems, not by RADIUS.

Conclusion

This lab demonstrated how FreeRADIUS can be used to implement a complete authentication and authorization workflow for a corporate wireless environment. By defining three distinct user groups—FTEs, Contractors, and Guests—and mapping their business requirements to specific RADIUS attributes, the system enforced identity-based network segmentation, time-of-day restrictions, and session-level controls. The configuration in the users file correctly applied VLAN assignments (10, 20, and 30), Login-Time restrictions for Contractors and Guests, and session timeouts aligned with organizational policy.

Testing with radtest validated the full lifecycle of the system. FTEs received unrestricted Access-Accept responses with VLAN 10; Contractors received Access-Accept responses during business hours and Access-Reject responses outside the allowed window; and Guests received limited Access-Accept responses with short session timeouts and VLAN 30. These results confirm that FreeRADIUS correctly enforced both authentication and authorization policies and that the configuration aligns with the scenario's security objectives.

In a real enterprise deployment, these RADIUS-level controls would be complemented by firewall rules, DLP systems, and identity governance policies to enforce restrictions on cloud storage and AI tool usage. Together, these layers form a cohesive Zero Trust architecture that limits lateral movement, reduces insider-risk exposure, and ensures that users receive only the access required for their role. This lab illustrates how FreeRADIUS serves as a foundational component of that architecture by providing reliable, role-based network access control.

References

Sawant, A. (2026). Authentication and Authorization. [Online course]. Coursera & Northeastern University.

Methodology and Use of AI Tools

I completed all technical steps of this lab directly in the Kali Linux virtual machine. All screenshots and observations were generated from my own live capture sessions. I used Microsoft Copilot as a writing and organizational aid to help structure the report, refine transitions between sections, and ensure alignment with the assignment rubric. I wrote up the scenarios and asked CoPilot to draft the configuration file based on my specifications. All analysis, interpretations, and conclusions reflect my own understanding of the material and the concepts covered in Module 9.

APPENDIX: Documented Process

The following table documents the exact steps performed for this lab, with each left-column description corresponding to the screenshot or output shown in the right column.

Step 1: Clone Kali Linux, update package list & install FreeRADIUS & Testing Utilities	<pre>***** Setting up freeradius (3.2.8+dfsg-1+b1) ... update-rc.d: We have no instructions for the freeradius init script. update-rc.d: It looks like a network service, we disable it. update-rc.d: warning: start and stop actions are no longer supported; falling back to defaults freeradius.service is a disabled or a static unit, not starting it. Setting up freeradius-utils (3.2.8+dfsg-1+b1) ... Processing triggers for man-db (2.13.1-1) ... Processing triggers for kali-menu (2025.4.3) ... (jenn@kali)-[~] \$ sudo nano /etc/freeradius/3.0/users</pre>
Step 2: Create testuser and restart FreeRADIUS in debug mode	View of debug mode: <pre> idle_timeout = 30 } } listen { type = "auth" ipv6addr = :: port = 0 limit { max_connections = 16 lifetime = 0 idle_timeout = 30 } } listen { type = "acct" ipv6addr = :: port = 0 limit { max_connections = 16 lifetime = 0 idle_timeout = 30 } } Listening on auth address 127.0.0.1 port 18120 bound to server inner-tunnel Listening on auth address * port 1812 bound to server default Listening on acct address * port 1813 bound to server default Listening on auth address :: port 1812 bound to server default Listening on acct address :: port 1813 bound to server default Listening on proxy address * port 37311 Listening on proxy address :: port 36009 Ready to process requests</pre>

Step 3: Test and show “Accept” message with correct password and Reject message with wrong password	<pre> jenn@kali: ~ Session Actions Edit View Help (jenn@kali)-[~] \$ radtest testuser password123 localhost 0 testing123 Sent Access-Request Id 247 from 0.0.0.0:55751 to 127.0.0.1:1812 length 78 User-Name = "testuser" User-Password = "password123" NAS-IP-Address = 127.0.1.1 NAS-Port = 0 Message-Authenticator = 0x00 Cleartext-Password = "password123" Received Access-Accept Id 247 from 127.0.0.1:1812 to 127.0.0.1:55751 length 38 Message-Authenticator = 0x86688ee34b46b163cbb062ec477f16e4 (jenn@kali)-[~] \$ radtest testuser password1236 localhost 0 testing123 Sent Access-Request Id 137 from 0.0.0.0:33184 to 127.0.0.1:1812 length 78 User-Name = "testuser" User-Password = "password1236" NAS-IP-Address = 127.0.1.1 NAS-Port = 0 Message-Authenticator = 0x00 Cleartext-Password = "password1236" Received Access-Reject Id 137 from 127.0.0.1:1812 to 127.0.0.1:33184 length 38 Message-Authenticator = 0x1e1344a462e1c7230f0b9e8b433998a0 (0) -: Expected Access-Accept got Access-Reject (jenn@kali)-[~] \$ </pre>
Step 4: create hashed password	<pre> (jenn@kali)-[~] \$ mkpasswd -m sha-512 password123 \$6\$otHJjQVIgLnVESJ.\$dEby2gr3Gy0KHfzLA5lpZlj502Df1NLSAu6sIJ6zJisdLP7IV2wEh75B1bL.L.yZQ9utqkarYFreurGmXNw48. (jenn@kali)-[~] \$ </pre>
Step 5: Restart FreeRADIUS and test again in other terminal window:	<pre> (jenn@kali)-[~] \$ radtest testuser password123 localhost 0 testing123 Sent Access-Request Id 203 from 0.0.0.0:33687 to 127.0.0.1:1812 length 78 User-Name = "testuser" User-Password = "password123" NAS-IP-Address = 127.0.1.1 NAS-Port = 0 Message-Authenticator = 0x00 Cleartext-Password = "password123" Received Access-Accept Id 203 from 127.0.0.1:1812 to 127.0.0.1:33687 length 38 Message-Authenticator = 0x2a7fd40fecbe240c78621e74c3f67892 </pre>
Step 6: Create a strong shared secret and update the configuration file	<pre> # # The default secret below is only for testing, and should # not be used in any real environment. # secret = 0uioXXC3H2lM/hxJF+mZcYor6Gd0oZvZlefN8e+0J+s= # # The global configuration "security.require_message_authenticator" # flag sets the default for all clients. That default can be # over-riden here, by setting it to a value. If no value is set, # then the default from the "radiusd.conf" file is used. # </pre>

Step 7: retest with the shared secret and confirm it works

```
(jenn@kali)-[~]
$ radtest testuser password123 localhost 0 0uioXXC3H2LM/hxJF+mZcYor6Gd0oZvZl
efN8e+0J)+s=
Sent Access-Request Id 81 from 0.0.0.0:35645 to 127.0.0.1:1812 length 78
  User-Name = "testuser"
  User-Password = "password123"
  NAS-IP-Address = 127.0.1.1
  NAS-Port = 0
  Message-Authenticator = 0x00
  Cleartext-Password = "password123"
Received Access-Accept Id 81 from 127.0.0.1:1812 to 127.0.0.1:35645 length 38
  Message-Authenticator = 0x52a2c4c007ef935624afaec49433daf3

(jenn@kali)-[~]
$
```

Step 8: add two users to users file and test both:

```
(jenn@kali)-[~]
$ radtest adminuser adminpass localhost 0 0uioXXC3H2LM/hxJF+mZcYor6Gd0oZvZl
efN8e+0J)+s=
Sent Access-Request Id 135 from 0.0.0.0:41624 to 127.0.0.1:1812 length 79
  User-Name = "adminuser"
  User-Password = "adminpass"
  NAS-IP-Address = 127.0.1.1
  NAS-Port = 0
  Message-Authenticator = 0x00
  Cleartext-Password = "adminpass"
Received Access-Accept Id 135 from 127.0.0.1:1812 to 127.0.0.1:41624 length 54
  Message-Authenticator = 0xa6fc8367263055c199e3255d2b6be974
  Tunnel-Type:0 = VLAN
  Tunnel-Medium-Type:0 = IEEE-802
  Tunnel-Private-Group-Id:0 = "10"

(jenn@kali)-[~]
$ radtest guestuser guestpass localhost 0 0uioXXC3H2LM/hxJF+mZcYor6Gd0oZvZl
efN8e+0J)+s=
Sent Access-Request Id 109 from 0.0.0.0:49869 to 127.0.0.1:1812 length 79
  User-Name = "guestuser"
  User-Password = "guestpass"
  NAS-IP-Address = 127.0.1.1
  NAS-Port = 0
  Message-Authenticator = 0x00
  Cleartext-Password = "guestpass"
Received Access-Accept Id 109 from 127.0.0.1:1812 to 127.0.0.1:49869 length 60
  Message-Authenticator = 0x05abd0d94931add3b405d37885803c42
  Session-Timeout = 1800
  Tunnel-Type:0 = VLAN
  Tunnel-Medium-Type:0 = IEEE-802
  Tunnel-Private-Group-Id:0 = "20"

(jenn@kali)-[~]
$
```

Step 9: Create simple authentication & authorization setup:

```
Session Actions Edit View Help
GNU nano 8.6 /etc/freeradius/3.0/users *
#####
# Full-Time Employee (FTE)
#####
fteuser Cleartext-Password := "ftepass"
Tunnel-Type = VLAN,
Tunnel-Medium-Type = IEEE-802,
Tunnel-Private-Group-Id = "10"

#####
# Contractor
#####
contractoruser Cleartext-Password := "contractorpass"
Session-Timeout = 3600,
Login-Time := "Wk0800-1800",
Tunnel-Type = VLAN,
Tunnel-Medium-Type = IEEE-802,
Tunnel-Private-Group-Id = "20"

#####
# Guest
#####
guestuser Cleartext-Password := "guestpass"
Session-Timeout = 1800,
Login-Time := "Wk0800-1800",
Tunnel-Type = VLAN,
Tunnel-Medium-Type = IEEE-802,
Tunnel-Private-Group-Id = "30"
```

Step 10: Test each role

```
(jenn@kali)-[~]
$ radtest fteuser ftepass localhost 0 0uioXXC3H2lM/hxJF+mZcYor6Gd0oZvZlefN8e+0J+s=
Sent Access-Request Id 34 from 0.0.0.0:46924 to 127.0.0.1:1812 length 77
  User-Name = "fteuser"
  User-Password = "ftepass"
  NAS-IP-Address = 127.0.1.1
  NAS-Port = 0
  Message-Authenticator = 0x00
  Cleartext-Password = "ftepass"
Received Access-Accept Id 34 from 127.0.0.1:1812 to 127.0.0.1:46924 length 54
  Message-Authenticator = 0x35eec30147163c24ee1c597dfcaa8a5b
  Tunnel-Type:0 = VLAN
  Tunnel-Medium-Type:0 = IEEE-802
  Tunnel-Private-Group-Id:0 = "10"

(jenn@kali)-[~]
$ radtest contractoruser contractorpass localhost 0 0uioXXC3H2lM/hxJF+mZcYor6Gd0oZvZlefN8e+0J+s=
Sent Access-Request Id 74 from 0.0.0.0:54645 to 127.0.0.1:1812 length 84
  User-Name = "contractoruser"
  User-Password = "contractorpass"
  NAS-IP-Address = 127.0.1.1
  NAS-Port = 0
  Message-Authenticator = 0x00
  Cleartext-Password = "contractorpass"
Received Access-Accept Id 74 from 127.0.0.1:1812 to 127.0.0.1:54645 length 60
  Message-Authenticator = 0x370ef0a9277afe0fd1c3cfd8967f6201
  Session-Timeout = 3600
  Tunnel-Type:0 = VLAN
  Tunnel-Medium-Type:0 = IEEE-802
  Tunnel-Private-Group-Id:0 = "20"
```

```
(jenn@kali)-[~]  
$ radtest guestuser guestpass localhost 0 0uioXXC3H2lM/hxJF+mZcYor6GdOoZvZl  
efN8e+0J+s=  
Sent Access-Request Id 141 from 0.0.0.0:34170 to 127.0.0.1:1812 length 79  
  User-Name = "guestuser"  
  User-Password = "guestpass"  
  NAS-IP-Address = 127.0.1.1  
  NAS-Port = 0  
  Message-Authenticator = 0x00  
  Cleartext-Password = "guestpass"  
Received Access-Accept Id 141 from 127.0.0.1:1812 to 127.0.0.1:34170 length 6  
0  
  Message-Authenticator = 0xc518e3b3d491a2043f018f7473983bb7  
  Session-Timeout = 1800  
  Tunnel-Type:0 = VLAN  
  Tunnel-Medium-Type:0 = IEEE-802  
  Tunnel-Private-Group-Id:0 = "30"
```

APPENDIX: Users File Configuration (/etc/freeradius/3.0/users)

```
#####  
# Full-Time Employee (FTE)  
#####  
fteuser Cleartext-Password := "ftepass"  
    Tunnel-Type = VLAN,  
    Tunnel-Medium-Type = IEEE-802,  
    Tunnel-Private-Group-Id = "10"  
  
#####  
# Contractor  
#####  
contractoruser Cleartext-Password := "contractorpass"  
    Session-Timeout = 3600,  
    Login-Time := "Wk0800-1800",  
    Tunnel-Type = VLAN,  
    Tunnel-Medium-Type = IEEE-802,  
    Tunnel-Private-Group-Id = "20"  
  
#####  
# Guest  
#####  
guestuser Cleartext-Password := "guestpass"  
    Session-Timeout = 1800,  
    Login-Time := "Wk0800-1800",  
    Tunnel-Type = VLAN,  
    Tunnel-Medium-Type = IEEE-802,  
    Tunnel-Private-Group-Id = "30"
```